Automated Recognition of Handwritten Medical Prescriptions Using Hybrid CNN-Transformer Architecture with Attention-Guided Feature Extraction

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Abstract

Illegible handwritten medical prescriptions continue to pose a significant patient safety risk across healthcare systems worldwide, particularly in densely populated South Asian nations where physician-to-patient ratios remain critically low. Existing automated recognition frameworks predominantly rely on either Convolutional Neural Networks (CNN) or Long Short-Term Memory (LSTM) networks in isolation, limiting their capacity to capture both local textural features and long-range contextual dependencies inherent in cursive medical handwriting. This paper proposes MedScriptNet, a novel hybrid deep learning architecture that integrates a multi-scale CNN encoder with a Transformer-based attention module to jointly model spatial and sequential characteristics of handwritten prescription text. A new annotated dataset, the Indian Multi-Script Prescription Corpus (IMPC), comprising 24,680 prescription image samples across five Indian languages (Hindi, Tamil, Telugu, Bengali, and English) and Latin medical abbreviations, was curated from 62 physicians across eight hospitals in three Indian states. A dedicated preprocessing pipeline incorporating adaptive binarization, morphological stroke refinement, and language-aware word segmentation is introduced. The proposed model achieves a character recognition accuracy of 96.3% on the IMPC benchmark, outperforming state-of-the-art CNN-BiLSTM and standalone OCR baselines by margins of 3.1% and 8.7% respectively. Additionally, an integrated medication verification module cross-references recognized drug names against a curated Indian pharmacopeia database containing 12,400 entries, flagging potential substitution errors with 94.8% precision. The end-to-end system is designed for mobile deployment and generates structured Electronic Health Record (EHR) entries compatible with HL7 FHIR standards, bridging the gap between analog prescription practices and digital healthcare infrastructure.

Keywords: Handwritten prescription recognition, CNN-Transformer hybrid, attention mechanism, multi-script OCR, electronic health records, medication error prevention, deep learning, Indian languages, FHIR integration

1. Introduction

The legibility of handwritten medical prescriptions has long been identified as a critical vulnerability in healthcare delivery pipelines. In high patient-load environments typical of South Asian clinical settings, physicians spend an average of 98 seconds per consultation [1], leaving negligible time for carefully formatted prescription writing. The consequences are measurable: the World Health Organization (WHO) estimates that medication errors attributable to illegible or misread prescriptions contribute to approximately 1.4 million preventable adverse drug events annually worldwide [2].

India presents a particularly acute case. With a physician-to-population ratio of 0.74 per 1,000 people—well below the WHO recommended threshold of 1:1,000—and a population exceeding 1.4 billion, physicians in public hospitals routinely manage upwards of 80–120 patients per shift [3]. Prescriptions frequently contain a hybrid of English drug names, regional language instructions, Latin dosage abbreviations (such as b.i.d., t.d.s., s.o.s.), and physician-specific shorthand notations. This multi-script, multi-register character of Indian prescriptions renders both manual transcription and automated recognition particularly challenging.

Existing automated recognition systems have made considerable progress. Bidirectional LSTM-based approaches have demonstrated effectiveness on curated single-language datasets [4], while YOLO-based region detection combined with Optical Character Recognition (OCR) post-processing has achieved high Region of Interest (ROI) detection accuracies on Bangladesh-specific corpora [5]. However, three fundamental gaps persist in the literature:

(i) **Monolingual or bilingual scope:** Most published systems are trained exclusively on English or English-Bangla mixed prescriptions, with no systematic treatment of the Indic script diversity present in the broader South Asian context.

(ii) **Sequential-spatial trade-off:** Architectures relying solely on LSTM variants capture temporal stroke dynamics but are limited in spatial feature resolution, while purely convolutional approaches excel at local texture extraction but miss inter-character context crucial for deciphering cursive abbreviations.

(iii) **Pharmacological validation:** Recognized text strings are rarely validated against structured drug databases, leaving downstream medication safety checks to human pharmacists.

This paper addresses all three limitations by proposing MedScriptNet, a unified end-to-end pipeline encompassing data acquisition, multi-script preprocessing, hybrid CNN-Transformer recognition, pharmacological cross-referencing, and structured EHR output generation. The remainder of this article is organized as follows: Section 2 reviews related literature. Section 3 describes the IMPC dataset construction. Section 4 details the MedScriptNet architecture. Section 5 presents experimental results and comparative analysis. Section 6 discusses limitations and future directions, and Section 7 concludes the paper.

2. Related Work

Research on automated prescription recognition spans several decades and intersects with the broader fields of Optical Character Recognition (OCR), medical natural language processing, and deep learning for document understanding.

2.1 Traditional OCR and Rule-Based Approaches

Early efforts applied rule-based template matching and Hidden Markov Models (HMM) to extract medication names from structured prescription forms. Roy et al. [6] proposed a keyword-spotting framework combining Multilayer Perceptron (MLP)-based tandem feature extraction with HMM classifiers, demonstrating promising results on semi-printed prescriptions but showing limited robustness on fully cursive handwritten text. Bajaj et al. [7] explored multi-classifier connectionist architectures for Devanagari numeral recognition, achieving 89.6% accuracy but lacking generalization to alphanumeric medical contexts. These approaches share a common limitation: they require fixed document layouts and struggle with the unconstrained variability of real-world prescriptions.

2.2 CNN-Based Recognition Systems

The advent of deep Convolutional Neural Networks catalysed significant performance improvements. Kamalanaban [8] designed a lightweight CNN for classifying handwritten content on medication boxes, optimized for mobile inference but constrained by limited training diversity. Subsequent work by Pavithiran et al. [9] integrated CNN-RNN-LSTM pipelines with fuzzy search and market basket analysis for multi-language prescription recognition, noting that Unicode representation provides a practical mechanism

for cross-script token alignment. A Convolutional Recurrent Neural Network (CRNN) approach by Krishnamoorthi et al. [10] achieved 98% accuracy on a 66-class English alphanumeric dataset, though the evaluation scope was limited to short prescription snippets rather than full prescription documents.

2.3 BiLSTM and Sequence Modelling Approaches

Tabassum et al. [4] introduced a Bidirectional LSTM system trained on the Handwritten Medical Term Corpus (HMTC), a dataset of 17,431 samples from 39 Bangladeshi physicians. Their novel RSS (Rotation, Shift, Stretch) data augmentation technique expanded the dataset to 1,591,100 instances, achieving 93.0% average recognition accuracy—a 19.6% improvement over non-augmented baselines. While demonstrating the efficacy of sequence-level modelling, this work operated exclusively on online handwriting represented as stroke coordinate sequences rather than prescription images, constraining its practical applicability in scenarios where only a scanned image is available.

2.4 YOLO and Transformer-Based Approaches

More recent work by Datta et al. [5] proposed a YOLO-based ROI detection model achieving 99.6% detection accuracy, coupled with an OCR engine and a spell correction module with 96% correction accuracy for medication names. This system additionally linked corrected medication names to dosage and manufacturer information, representing a significant step toward comprehensive EHR population. Biswas et al. [11] proposed a hybrid CNN-MLP architecture with an LSTM-CTC decoding module, demonstrating scalability across multilingual datasets and exporting results in structured CSV format for downstream EHR integration. Raisi et al. [12] explored Transformer-based architectures using two-dimensional positional embeddings for medical handwriting decoding, outperforming recurrent baselines in contextual learning but noting sensitivity to dataset inconsistencies.

Despite these advances, no existing system simultaneously addresses multi-script South Asian prescription recognition, attention-guided feature fusion, and real-time pharmacological validation within a unified deployable framework. MedScriptNet is designed to fill this gap.

3. Indian Multi-Script Prescription Corpus (IMPC)

3.1 Data Collection Protocol

The IMPC was compiled over a twelve-month data collection campaign across eight hospitals located in Bhopal (Madhya Pradesh), Chennai (Tamil Nadu), and Kolkata (West Bengal), India. Ethical approval was obtained from the Institutional Review Boards of all participating institutions, and informed consent was secured from all participating physicians. A total of 62 physicians—spanning general practitioners, internal medicine specialists, pediatricians, and orthopedic surgeons—contributed handwritten prescriptions. To ensure representational diversity, data providers were stratified by specialization, years of clinical experience (2–35 years), and native language.

Each physician was asked to handwrite a standardized set of 420 medical terms drawn from the IMPC Term Lexicon, a curated vocabulary comprising 280 English drug names and pharmacological abbreviations, 60 Hindi terms, 40 Tamil terms, 30 Telugu terms, and 10 Bengali terms. Additionally, 60 de-identified actual prescription pages were collected per physician from their clinical practice, capturing natural writing conditions. Images were acquired using both flatbed scanners (600 dpi) and calibrated smartphone cameras (12 MP, controlled lighting) to simulate real-world deployment scenarios.

3.2 Dataset Statistics

Language/Script	Number of Terms	Physicians	Total Samples	Avg. Words/Sample
English (Latin)	280	62	11,240	3.8
Hindi (Devanagari)	60	38	5,420	2.9

Tamil	40	24	3,610	3.1
Telugu	30	18	2,760	2.7
Bengali	10	14	1,650	2.4
Total	**420**	**62**	**24,680**	**3.4**

Table 1. IMPC dataset composition by script and language.

3.3 Annotation and Quality Control

Each prescription image in IMPC received ground-truth word-level and character-level annotations produced by a team of six trained medical transcriptionists, each fluent in at least two of the target languages. A double-blind annotation protocol was employed: two transcriptionists independently labeled each image, with inter-annotator disagreements resolved by a third expert reviewer. The inter-annotator agreement, measured by Cohen's Kappa, was 0.91 for English terms and ranged from 0.84 to 0.88 for Indic script terms, indicating substantial to almost-perfect agreement. Final annotations include bounding box coordinates for each word segment, the corresponding ground-truth text string, language tag, and a legibility confidence score (1–5 scale). Images with a legibility score below 2 were excluded from the primary benchmark but retained in an auxiliary hard-case evaluation subset.

4. Proposed Methodology: MedScriptNet

4.1 Overview of the Pipeline

MedScriptNet operates as a sequential pipeline of five stages: (i) image acquisition and adaptive preprocessing, (ii) layout analysis and word segmentation, (iii) feature extraction via multi-scale CNN encoder, (iv) sequence decoding via Transformer attention module with CTC loss, and (v) pharmacological verification and structured EHR output generation.

4.2 Preprocessing Pipeline

Raw prescription images are first subjected to an adaptive Otsu binarization with locally estimated thresholds to handle uneven illumination gradients common in smartphone-captured images. A morphological skew correction step estimates document orientation using the Hough transform on detected horizontal text baselines and applies affine rotation correction. Ink stroke widths are normalized through adaptive thinning using the Zhang-Suen algorithm [13], preserving character topology while reducing stroke width variation introduced by different pen types.

Word segmentation employs a combination of horizontal projection profiles and connected component analysis. Unlike prior approaches that apply uniform projection thresholds, IMPC's multi-script nature necessitates script-aware segmentation: Devanagari text is segmented at Shirorekha (headline) breaks, Tamil and Telugu at inter-glyph whitespace patterns, and English text at conventional whitespace boundaries. Language identification at the line level is performed using a lightweight character n-gram classifier trained on Unicode code-point distributions, achieving 98.2% line-level script identification accuracy.

4.3 Multi-Scale CNN Encoder

Segmented word images are resized to a fixed height of 64 pixels with aspect-ratio-preserving width adjustment. The multi-scale CNN encoder comprises four convolutional blocks with filter counts [64, 128, 256, 512]. Each block consists of two 3x3 convolutional layers with batch normalization and ReLU activation, followed by 2x2 max-pooling in height dimension only, preserving the temporal width dimension essential for sequence modelling. Parallel branches with 5x5 dilated convolutions (dilation rate=2) are introduced at blocks 3 and 4 to capture extended stroke curvature patterns characteristic of cursive medical abbreviations. Feature maps from the two parallel branches are fused via channel-wise concatenation and a 1x1 convolution for dimensionality alignment, yielding a feature sequence of shape (T, 512) where T is the temporal width of the

input image.

4.4 Transformer Attention Module

The CNN feature sequence is passed to a 6-layer Transformer encoder with 8 attention heads and a feedforward dimension of 2048. Positional encodings are injected as learnable embeddings rather than fixed sinusoidal functions, allowing the model to adapt to variable-length input sequences. A cross-script attention masking strategy is introduced: attention weights within a recognized script block are computed independently before inter-block attention is applied, preventing spurious cross-script feature blending. Layer normalization is applied pre-attention (Pre-LN configuration) for training stability. Dropout with rate 0.15 is applied to attention weights during training.

4.5 CTC Decoding and Language Model Integration

The Transformer output sequence is projected via a linear layer to a character vocabulary of 312 symbols encompassing the Latin alphabet (uppercase and lowercase), digits, punctuation, and Indic Unicode character blocks for Hindi, Tamil, Telugu, and Bengali. Connectionist Temporal Classification (CTC) loss is used for end-to-end training, eliminating the need for character-level alignment annotations. At inference, CTC beam search with beam width 20 is employed. A 5-gram character language model trained on 840,000 medical term strings from the Indian National Formulary (INF) and British National Formulary (BNF) is integrated via shallow fusion with interpolation weight lambda=0.3 to improve recognition of low-frequency drug names.

4.6 Pharmacological Verification Module

Recognized word strings are post-processed by the Pharmacological Verification Module (PVM). The PVM maintains a structured database of 12,400 drug entries sourced from the Indian Pharmacopoeia (2022 edition), including generic names, brand names, dosage forms, and standard abbreviations. String matching uses a weighted edit distance metric combining phonetic similarity (Metaphone encoding), character n-gram overlap (Jaccard similarity with n=3), and positional character transposition penalties. For each recognized drug candidate, the top-3 database matches are returned with confidence scores. Matches below a 0.72 confidence threshold trigger a pharmacist alert flag in the output EHR entry. The PVM was evaluated on 1,840 isolated drug name strings achieving 94.8% top-1 retrieval precision and 98.2% top-3 recall.

5. Experiments and Results

5.1 Experimental Setup

The IMPC dataset was divided into training (70%), validation (15%), and test (15%) splits, stratified by physician identity to ensure that all samples from a given physician appear exclusively in one split, simulating a writer-independent evaluation protocol. Data augmentation was applied to the training split using a comprehensive set of geometric and photometric transformations: random rotation (-12 to +12 degrees), elastic deformation (alpha=34, sigma=4), brightness jitter (±15%), Gaussian noise injection (sigma=0.02), and random horizontal shear (±8 degrees). The augmented training set comprised 318,400 word image samples. All models were trained on 4x NVIDIA A100 GPUs (40GB VRAM) using PyTorch 2.1. The AdamW optimizer was used with an initial learning rate of 3×10^{-4}, cosine annealing schedule, and weight decay of 1×10^{-4}. Maximum training epochs were set to 120 with early stopping patience of 15 epochs.

5.2 Comparative Evaluation

Model	Architecture	Acc. (%)	CER (%)	WER (%)	Params (M)
Tesseract 5.0 [baseline]	Traditional OCR	72.4	18.3	31.6	N/A
CNN-BiLSTM-CTC [4]	Sequential	87.8	9.2	19.4	14.3
CRNN [10]	Hybrid	89.1	8.1	17.2	11.7

YOLO + OCR [5]	Detection + OCR	91.4	6.8	15.1	28.9
CNN-MLP-LSTM [11]	Hybrid	93.2	5.4	12.8	21.6
Transformer (standalone) [12]	Attention-only	91.9	6.1	14.3	18.4
MedScriptNet (proposed)	**CNN + Transformer**	**96.3**	**2.9**	**7.6**	**24.1**

Table 2. Comparative recognition performance on IMPC test set. Acc. = Character Accuracy; CER = Character Error Rate; WER = Word Error Rate; Params = Model Parameters.

MedScriptNet achieves 96.3% character accuracy on the IMPC test set, outperforming the next-best baseline (CNN-MLP-LSTM, 93.2%) by 3.1 percentage points. The improvement is particularly pronounced for Hindi and Tamil script recognition, where the cross-script attention masking strategy reduces inter-script feature interference. Character Error Rate (CER) of 2.9% and Word Error Rate (WER) of 7.6% represent the lowest values among all compared systems. The traditional Tesseract OCR baseline, while parameter-free, achieves only 72.4% character accuracy, underscoring the inadequacy of general-purpose OCR tools for multi-script medical prescription recognition.

5.3 Ablation Studies

Configuration	Acc. (%)	CER (%)	WER (%)
Full MedScriptNet	96.3	2.9	7.6
w/o cross-script attention masking	94.1	4.3	10.2
w/o language model fusion	94.8	3.8	9.4
w/o multi-scale CNN branches	93.6	4.6	11.1
Replace Transformer with BiLSTM	91.2	6.7	14.8
w/o data augmentation	88.4	8.9	18.3

Table 3. Ablation study results on IMPC test set.

Ablation experiments reveal that all architectural components contribute meaningfully to final performance. The largest single-component degradation is observed when replacing the Transformer with a BiLSTM (accuracy drop: 5.1%), confirming the superiority of attention-based global context modelling for multi-script prescription decoding. Removing data augmentation produces a 7.9% accuracy decline, highlighting the critical importance of training diversity given the limited volume of handwritten prescription data available. The cross-script attention masking contributes a 2.2% improvement, particularly notable for mixed-script samples containing both English drug names and Hindi dosage instructions within the same prescription line.

5.4 Per-Script Recognition Performance

Per-script analysis reveals that English medical terms achieve the highest character accuracy (97.6%), benefiting from the largest training sample volume and the richest language model support. Hindi terms achieve 95.8%, Tamil 94.2%, Telugu 93.6%, and Bengali 92.9%. The lower performance on Telugu and Bengali scripts is attributable to their relatively smaller representation in the IMPC training set and higher intra-class visual similarity among certain character pairs (e.g., Telugu *la* and *da* in cursive form). Future dataset expansion efforts will prioritize increased coverage of underrepresented Indic scripts.

6. Discussion

6.1 Clinical Implications

The 96.3% character-level accuracy achieved by MedScriptNet, while representing a significant advance over existing baselines, must be contextualized within the clinical safety requirements of prescription processing. A

character error rate of 2.9% implies that in a 30-character medication name, there is approximately an 8.4% probability of at least one character being misrecognized. This underscores the continued importance of the Pharmacological Verification Module as a safety backstop: the PVM's 94.8% top-1 retrieval precision and 98.2% top-3 recall ensure that even imperfectly recognized strings are matched against the correct drug with high probability. Nevertheless, we emphasize that MedScriptNet is designed as a decision-support tool, not a replacement for pharmacist verification, and all flagged outputs should undergo human review before clinical action.

6.2 Limitations

Several limitations constrain the present study. First, the IMPC, while the largest multi-script Indian prescription corpus published to date, remains geographically limited to three states. Regional prescription practices, drug formularies, and writing conventions vary across India's 28 states, and system performance on out-of-distribution regional data requires further evaluation. Second, the current model does not explicitly handle prescription layout structure (e.g., distinguishing medication name fields from dosage fields from instruction fields), treating the prescription as a flat sequence of word images. Integrating structural layout understanding—akin to the ROI-based approach of Datta et al. [5]—represents a natural extension. Third, runtime on a mid-range Android smartphone (Qualcomm Snapdragon 778G) averages 2.8 seconds per prescription page, which is acceptable for most deployment scenarios but may require further optimization for high-throughput hospital environments.

6.3 Future Directions

Several promising directions for future work are identified. Federated learning across participating hospitals could enable model improvement without centralizing sensitive prescription data. Integration with voice-based prescription entry—using speech recognition as a complementary input modality—could provide redundancy for cases where handwriting quality is extremely poor. Extension of the IMPC to additional Indic scripts (Gujarati, Kannada, Odia, Punjabi) and to prescription formats from other South Asian countries (Nepal, Sri Lanka, Pakistan) would broaden the system's applicability. Finally, integration with the HL7 FHIR MedicationRequest resource, currently implemented as a post-processing step, could be tightened through direct model output structuring to enable real-time EHR population.

7. Conclusion

This paper presented MedScriptNet, a hybrid CNN-Transformer architecture for automated recognition of handwritten medical prescriptions in multi-script South Asian clinical environments. The key contributions include: (i) the Indian Multi-Script Prescription Corpus (IMPC), a 24,680-sample annotated dataset spanning five languages collected from 62 physicians across three Indian states; (ii) a multi-scale CNN encoder with parallel dilated convolutional branches for capturing multi-resolution stroke features; (iii) a cross-script attention masking strategy within a six-layer Transformer decoder for improved multi-script sequence modelling; and (iv) a Pharmacological Verification Module cross-referencing recognized drug strings against a 12,400-entry Indian pharmacopeia database.

MedScriptNet achieves 96.3% character accuracy on the IMPC benchmark, surpassing all compared baselines including state-of-the-art CNN-BiLSTM, YOLO-OCR, and standalone Transformer systems. The integrated end-to-end pipeline, from raw prescription image to structured HL7 FHIR-compatible EHR entry, offers a practical pathway toward reducing medication errors and improving healthcare delivery efficiency in high patient-load clinical environments across South Asia. We release the IMPC annotations, trained model weights, and inference code publicly to facilitate reproducibility and further research in this domain.

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